

Toshiba 1.5T Titan Monthly PM Report



Document #: 20160826

Revision: B

Affected Products

Toshiba 1.5T Titan

Reference Material

Vantage Titan Maintenance & Inspection Manual 2y900-045EN*x (or equivalent)

History

Date	Name	Rev	Change Descriptions
10/22/15	Document Control	A	Initial release of BCT format document
12/6/16	Document Control	B	Converted to standard document, new doc #

Approval

Department	Name	Date
Specialist	Brian Weindling	
Document Control	Jhovany Hernandez	12/6/16

Preface

This document is valid proof of a Preventative Maintenance performed on the system. The Preventative Maintenance must be performed in the specified intervals, and the results are to be documented herein.

Refer to **Reference Material listed above** for specific procedures.

This section addresses planned maintenance that should be performed by service personnel. Some of it is also listed in the Operator's Manual for the user to perform on daily and weekly basis.

However, service personnel should make sure that these procedures have been followed and if not, include them on the periodic visits.

Warning: All steps must be performed in accordance with the manufacturer's service manual.

This document must be completely filled out by the Field Service Engineer, i.e.:

- All boxes must be filled out. Use N/A where appropriate.
- Enter the Serial Number and the Date in the header of each page.
- After completing the check, fill out and sign the last page.

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System Information

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Customer: Operator Reported Problems

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FSE: Engineer Reported Problems

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Serial #:

Date:

Checklist

Perform the following steps and check off each step for the corresponding month when completed. Input any other data when requested. Each scanner is different. Consult the system's PM manual to identify the exact breakdown of each sub-section. Note the PM manual assigns tasks based on quarters. Assure that all required tasks are completed by the end of the assigned time period.

System Checks

<i>Image Quality</i>				Quarter (1-4):	
				Month (1-12):	
Pass	Fail	N/A	Action	Notes	Measurements/Comments
☐	☐	☐	Review system performance and image quality. Check with Tech/Radiologist.		

<i>Magnet Room</i>				Quarter (1-4):	
				Month (1-12):	
Pass	Fail	N/A	Action	Notes	Measurements/Comments
☐	☐	☐	Check magnet area for loose ferrous materials.		
☐	☐	☐	Check scan room door seal.	If damaged, notify customer.	
☐	☐	☐	Record magnet room temperature and humidity.	Degrees F (60-75):	
				% Humidity (40-50%):	
☐	☐	☐	Check magnet room lights.	Notify customer of any burnt out bulb.	
☐	☐	☐	Check magnet bore lighting.		
☐	☐	☐	Visually check coil port connectors.	Look for damage/debris. Clean.	
☐	☐	☐	Check table operation (in/out, up/down)	Verify proper operation.	
☐	☐	☐	Check gantry cooling fan operation.	No abnormal noises.	
☐	☐	☐	Clean bed/console area.	Wear gloves.	
☐	☐	☐	Check table manual brake release.	Verify lever is in middle position.	
☐	☐	☐	Check laser light operation/accuracy.	Adjust as necessary.	

<i>Coil QA</i>				Quarter (1-4):	
				Month (1-12):	
Pass	Fail	N/A	Action	Notes	Measurements/Comments
☐	☐	☐	Perform QDWBC DQA. Record signal/noise values.	Mean S/N SD. (Refer to appropriate coil install manual, 2C9xx-xxx)	

Serial #:

Date:

System Checks (Continued)

<i>Equipment Room</i>				Quarter (1-4):	
				Month (1-12):	
Pass	Fail	N/A	Action	Notes	Measurements/Comments
☐	☐	☐	Record equipment room temperature and humidity.	Degrees F (60-75):	
				% Humidity (40-50%):	

<i>Cooling Water Checks</i>				Quarter (1-4):	
				Month (1-12):	
Pass	Fail	N/A	Action	Notes	Measurements/Comments
☐	☐	☐	Check for water leaks temperature.		
☐	☐	☐	Rydlyme flush/refill.	Perform as necessary.	
☐	☐	☐	Check water reservoir level.		
☐	☐	☐	Cooling water manifold input pressure.	Psi/MPa	
☐	☐	☐	Cooling water temperature.	Degrees F (64-71):	
☐	☐	☐	Cooling water flow rate gradient coil .	GPM (>3.1 GPM):	
☐	☐	☐	Cooling water flow rate compressor .	GPM (1.6-2.4 GPM):	
☐	☐	☐	Cooling water flow rate gradient cabinet .	GPM (>8.0 GPM):	
☐	☐	☐	Replace gradient cabinet water filter.	Every 3 months.	
☐	☐	☐	Replace compressor output water filter.	Every 3 months.	
☐	☐	☐	Check for visible air in flow gauges.	Purge any air.	

<i>Gradient Coil Flow Switch</i>				Quarter (1-4):	
				Month (1-12):	
Pass	Fail	N/A	Action	Notes	Measurements/Comments
☐	☐	☐	Remove water flow to g-coil. Confirm error occurs and system cannot be operated.	System error must occur.	
☐	☐	☐	Check the inside of the flow switch. Confirm there is no blockage due to foreign matter.	Clean as necessary.	

Serial #:

Date:

System Checks (Continued)

<i>Vacuum Pump</i>				Quarter (1-4):		
				Month (1-12):		
Pass	Fail	N/A	Action	Notes	Measurements/Comments	
☐	☐	☐	Record vacuum level.	Kpa:		
☐	☐	☐	Check for water at vacuum pump output.	Clean as necessary.		

<i>Temperature Checks</i>				Quarter (1-4):		
				Month (1-12):		
Pass	Fail	N/A	Action	Notes	Measurements/Comments	
☐	☐	☐	Run Gif temp. Record temperatures.	OLP1:		
				OLP2:		
				OLP3:		
				Gantry:		
				BORE:		
				Rear Bird:		
				WBC:		

<i>Compressor</i>				Quarter (1-4):		
				Month (1-12):		
Pass	Fail	N/A	Action	Notes	Measurements/Comments	
☐	☐	☐	Check dynamic pressure (on).	bar (HC10L 20.0-22.8 bar):		
☐	☐	☐	Check static pressure (off).	bar (HC10L 20.0-22.8 bar):		
☐	☐	☐	Record hours.	Hrs:		
☐	☐	☐	Check for leaks/unusual noises.			

<i>Magnet</i>				Quarter (1-4):		
				Month (1-12):		
Pass	Fail	N/A	Action	Notes	Measurements/Comments	
☐	☐	☐	Record magnet pressure as measured on the pressure gauge.	PSIA/PSI:		
☐	☐	☐	Check helium level. Calculate 30-day LHe boil-off. (previous level - current level) / # of days	Previous LHe level %:		
				Date previous:		
				Current LHe level %:		
				Date current:		
				LHe level %/day:		
☐	☐	☐	Listen to coldhead for unusual knocking.	Report any unusual sound.		

Serial #:

Date:

System Condition

After performing scheduled PM, an evaluation of the system is performed with the following result (check one):

☐

No deficiencies. The image quality test resulted in no differences from required reference values.

☐

Slight deficiencies. There is no effect on operation; however they should be corrected preventively. The image quality test resulted in no differences from required reference values.

☐

Serious deficiencies. This must be corrected before operation of the system is permitted.

Field Service Engineer

Name:

Date:

Notes (Additional work to be performed, partial PM, customer requests, etc.):

Customer

Name:

Date: